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Missing Surface Inflection or Impairment in second language acquisition? Evidence from tense and agreement

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In this article, two accounts of the variable use of inflection in adult second language (L2) acquisition are examined. The Missing Surface Inflection Hypothesis (MSIH) proposes that L2 learners have unconscious knowledge of the functional projections and features underlying tense and agreement. However, learners sometimes have a problem with realization of surface morphology, such that they resort to non-finite forms (e.g. Haznedar and Schwartz, 1997; Prévost and White, 1999). The Impaired Representation Hypothesis (IRH) claims that L2 inflection is essentially impaired, due to lack of functional categories, features or feature strength (e.g. Eubank, 1993/94; Meisel, 1997). These views make different predictions for adult L2 acquisition. Spontaneous production data from two adult learners of French and two adult learners of German are examined. The data show that finite forms do not occur in non-finite contexts, that learners exhibit syntactic reflexes of finiteness and that inflected forms largely show accurate agreement. These results suggest that adult L2 learners represent finiteness and agreement at an abstract level, rather than being impaired in this domain, supporting the MSIH.

I Introduction

It is well known that second language (L2) learners exhibit optionality in their use of inflectional morphology, with tense and agreement markings sometimes being present and sometimes absent in L2 production data. This is true during the course of acquisition, as shown, for example, by the research on morpheme acquisition orders conducted during the 1970s (e.g. Dulay and Burt, 1974; Bailey *et al.*, 1974; for relevant discussion, see Zobl and Liceras, 1994). In some cases, variable use of inflectional morphology is found even in the endstate grammar (Lardiere,

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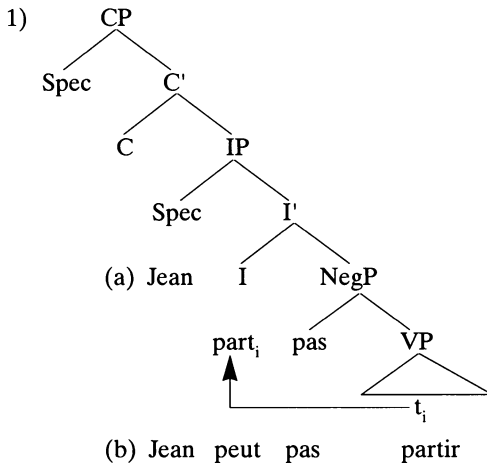
1998a; 1998b). While the fact of variation is uncontroversial, there is relatively little agreement as to what such optionality implies and, in particular, whether it indicates major impairment to the interlanguage grammar (ILG) in the domain of functional categories or whether it is the consequence of some more superficial problem.

In this article, we examine variability in use of verbal inflection in L2 French and German, considering the incidence of finite versus non-finite morphology and the contexts in which each is found, as well as verbal agreement. Following earlier work (Lardiere, 1998a; 1998b; Prévost and White, 1999), we argue that variability in adult L2 performance does not reflect a deeper lack of functional categories or features associated with tense and agreement. Rather, L2 learners have difficulties with the overt realization of morphology.

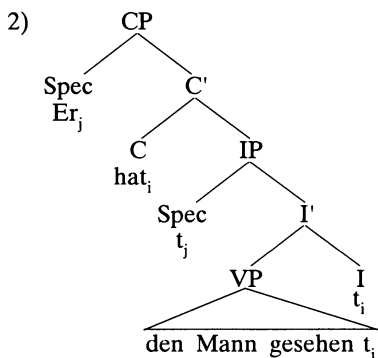
II Functional categories

The properties that we are interested in relate to verbal functional projections. Verb (V) features have to be checked in Infl (I) (Chomsky 1995). If V features are strong, the verb raises overtly to I for feature-checking purposes. If V features are weak, the verb raises covertly (after spell out). A number of syntactic consequences result: in languages like French and German, the L2s being considered here, V features are strong; finite verbs raise, resulting in well-known alternations between the positions of finite and non-finite verbs.

The tree in (1) illustrates V raising in French. The finite verb raises to I, over negation or adverbs (as in (1a)), in contrast to the non-finite verb, which remains in the VP (1b). Thus, finite verbs in French appear to the left of negation or adverbs, whereas non-finite verbs appear to the right (Pollock, 1989).



German is V final, as well as I final on many analyses (den Besten, 1983). In main clauses, the finite verb raises through I to C, as illustrated in (2), while some other phrase raises to Spec CP, resulting in the verb second (V2) effect. Nonfinite verbs, on the other hand, remain in final position. As in French, then, there are overt syntactic consequences of the $\pm$ finiteness distinction, such as V2 vs. V final, as well as the same alternation with respect to negation as is found in French.



While there is considerable agreement on the syntactic consequences of feature strength, namely presence or absence of verb raising, there is less consensus regarding the morphology. On the one hand there are those who argue for a very close connection between morphological paradigms and feature strength, such that absence of a paradigm implies absence of the relevant functional projections or feature strength (e.g., Clahsen *et al.*, 1996). On the other hand, there are those who argue that abstract featural

properties cannot be closely associated with morphological paradigms, given that there are languages which exhibit the syntactic reflexes of strong features without having rich morphology (Sprouse, 1998).

III Variability in L1 acquisition

One kind of variation in the incidence of verbal inflection which has been much investigated in first language (L1) acquisition is the Optional Infinitive (OI) (Wexler, 1994) or Root Infinitive (Rizzi, 1993/94) phenomenon, whereby children's main clauses vary between having a finite or a non-finite main verb. Although accounts of what is going on differ considerably, there is general agreement that OIs are structurally determined: when the child uses non-finite forms in place of finite ones, these are indeed non-finite and are found in structural positions typical of non-finite verbs. For example, the child's non-finite verbs do not raise past negation in French and are not found in the V2 position in German. When the child uses finite forms, on the other hand, these are found in structural positions typical of finite verbs. Variation in the use of finiteness, then, is not random.

In earlier work, Wexler (1994) argued that the OI stage can be accounted for on the assumption that Tense is underspecified in early child grammars; Hoekstra *et al.* (1997) argue that it is Number that is underspecified. On these accounts, functional categories in the grammars of L1 acquirers are represented in a manner that differs from the adult grammar. Rizzi (1994) and Haegeman (1995) offer a somewhat different analysis, involving truncation. They argue that early grammars possess the same set of functional categories as adult systems, and that none of them are underspecified. However, there is no restriction on what projection can serve as the root of a matrix declarative. The child's structure may be truncated at any point below CP, such that root VPs, root IPs, and root NegPs are possible, in addition to CPs. Variability, then, is the consequence of different roots being projected for different utterances. Nevertheless, incidence of finiteness is structurally constrained: root VPs are non-finite, whereas other roots are finite. Both the underspecification theory and the truncation theory assume that non-finite forms are indeed non-finite in L1 acquisition, showing the expected syntactic properties of non-finite verbs with respect to verb placement. The issue is not that the child does not know the difference between finite and non-finite forms; rather, optionality is the result of the child's not knowing that main clauses must be finite (on the OI account) or

that they must be CPs (on the truncation account).

A number of researchers have suggested that optionality in incidence of finite forms in L1 acquisition reflects a problem with realization of morphology, rather than any problem with functional categories or features as such (Borer and Rohrbacher, 1997; Boser *et al.*, 1992). On this approach, clauses containing non-finite lexical verbs are assumed to be finite (contra Wexler and Rizzi), containing a finite auxiliary which is null.¹ Again, non-finite forms occur in the positions expected of non-finite verbs; in other words, they do not raise.

The above accounts concentrate on the issue of finiteness. However, they also have something to say about the realization of agreement morphology. Borer and Rohrbacher (1997: 26) argue that abstract functional projections can exist in the grammar in the absence of overt morphology and propose that 'the absence of overt agreement markers . . . appears to be motivated by the desire to avoid incorrect forms whose morpho-phonology has not been fully acquired'. Poeppel and Wexler (1993) show that, while agreement morphology is often lacking in early L1 acquisition of German, when agreement is present it is accurate; for example, if a verb occurs with 3rd person singular morphology, the subject will be 3rd person singular. (As they point out, the reverse contingency does not hold, since a 3rd person singular subject may also be found with a verb form not marked for agreement.)

To summarize, while disagreeing over whether or not child grammars allow non-finite main clauses, the above accounts of variability in L1 acquisition agree that:

- 1) in the case of tense morphology, there may be variation between finite and non-finite forms but if a verb form is finite, it has finite properties, whereas if a form is non-finite, it has non-finite properties;
- 2) in the case of agreement morphology, variation is between forms being present and appropriate, or being absent; 'faulty' agreement is argued not to be found (e.g. 3rd person morphology with 2nd person subject).

We turn now to a consideration of variability in L2 acquisition.

IV Variability in L2 acquisition

Superficially, at least, L2 acquisition presents a similar picture, in that incidence of tense and agreement morphology is variable. At

¹ Radford (1992) also argues for a null finite Aux, once functional categories have emerged in the child grammar.

issue is whether the variability takes the same form as in L1 acquisition or whether it is of a different nature, and, if so, whether this reflects some form of permanent 'impairment' to the ILG.

Broadly speaking, there are two kinds of accounts of variable use of inflectional morphology in L2. On the one hand, there are researchers who propose that ILGs contain abstract functional categories and features, and that they show syntactic consequences of these properties, such as verb raising, case-assignment, presence or absence of null subjects, etc. Variability in morphology is argued to reflect difficulties in identifying the appropriate morphological realization of functional categories; that is, the problem lies in mapping from abstract features to their surface morphological manifestation (Epstein *et al.*, 1996; Grondin and White, 1996; Haznedar and Schwartz, 1997; Lardiere and Schwartz, 1997; Lardiere, 1998a, 1998b, 2000; Prévost and White, 1999). On the other hand are those who argue that L2 grammars suffer from some form of impairment in the functional domain, resulting in variability, either temporarily or permanently (Meisel, 1991, Eubank, 1993/94; Beck, 1998; Eubank *et al.*, 1997; Meisel, 1997; Beck, 1998).

1 Missing surface inflection

Following Haznedar and Schwartz (1997), we refer to the first approach as the Missing Inflection Hypothesis, emending this term to Missing Surface Inflection Hypothesis (MSIH) to make it clear that it is at the surface morphological level that inflection is assumed to be absent, rather than at the abstract featural level. This position has been advanced independently by a number of researchers; some have simply suggested that it must be the case without pursuing the issue in detail (e.g., Epstein *et al.*, 1996; Grondin and White, 1996). Recently, researchers have begun to investigate the properties of missing surface inflection in some detail (Haznedar and Schwartz, 1997; Lardiere, 1998a; 1998b; Martohardjono *et al.*, 2000; Prévost and White, 1999).

Arguing against an OI phenomenon in L2 acquisition, Haznedar and Schwartz (1997) looked at data from a Turkish child learning English as L2. While they found evidence of use of both finite and non-finite verb forms, they found little evidence of other properties that are normally associated with the OI stage, such as incidence of null subjects and accusative case in place of nominative. They also found that agreement was accurate when it occurred (as was also observed by Grondin and White (1996) for child L2 French). Haznedar and Schwartz argue that there is no evidence for a syntactic deficit, such as underspecification of Tense, in their

subject's English; rather, the child's non-finite morphology was indicative of missing inflection. They speculate as to whether the same phenomenon might be found in adult L2 acquisition.

Lardiere (1998a; 1998b) examined the L2 English of an adult Chinese-speaker, Patty, who had resided in the USA for 18 years and whose grammar is clearly at its end state. Lardiere shows that incidence of tense morphology in spontaneous production is at about 35%, while 3rd person singular agreement is less than 17% (excluding the copula and auxiliary *be*). In spite of this low level of overt morphology, Patty shows a variety of syntactic phenomena which suggest that she has tense and agreement, with appropriate weak values, at an abstract level. For example, Patty shows 100% correct incidence of nominative case assignment and complete knowledge of the fact that English verbs do not raise, suggesting that her problems are in the domain of morphological mapping rather than being due to any deficit in functional features as such.

Prévost (1997) and Prévost and White (1999) looked for evidence of truncation in L2 acquisition but found this only in the case of child L2 learners. The behaviour of adults was more consistent with missing surface inflection. To summarize their findings, co-occurrence of finite and non-finite main verbs was found in both the child and the adult data, suggesting variability. In the child L2 data, non-finite forms occurred in different contexts from finite forms; the distribution of non-finite verbs was structurally determined and consistent with truncation. The adult data, on the other hand, showed finite and non-finite verbs occurring in the same contexts (in CPs, with subject clitics, preceding negative markers, etc.). Many more (apparently) non-finite verbs were found in the adult corpora and the adults produced these forms throughout the period of observation.

All the accounts of missing L2 inflection described above suggest that non-finite or bare stems are being used as substitutes for finite forms. In such cases, then, main verbs bearing the infinitival marker are not in fact non-finite but are being used as default forms exhibiting properties of finite verbs. This assumption differs crucially from most accounts of variation in L1 acquisition, where non-finite forms are argued to be truly non-finite.²

2 Impairment

The accounts assuming missing surface inflection contrast with views that argue for some kind of impairment to the ILG, either

² But see Phillips (1996) for a different view: he assumes that non-finite main verbs in L1 acquisition are in fact finite but that they do not raise.

globally or locally, at the level of functional categories or features. We will refer to this position as the Impaired Representation Hypothesis (IRH). Meisel (1991; 1997) effectively assumes global impairment. He argues that L2 grammars lack agreement (on the grounds that infinitives and bare stems are common in place of forms appropriately inflected for agreement) and that finiteness and verb-placement are unrelated (see also Clahsen, 1988). Meisel (1997) found that non-finite forms occur in finite positions, in particular in the V2 position in L2 German. However, if finiteness and verb placement are truly unrelated, we would expect not only to find non-finite forms in finite positions but also finite forms in non-finite positions. Meisel argues that this is indeed the case for two of the three adults that he studied; in their L2 German negatives, the negator appears to the left of finite verbs and modals in some cases. He concludes that adult L2 acquisition is substantially different from child L1 acquisition. In particular, Universal Grammar (UG) is no longer involved; rather the adult proceeds in terms of linear sequencing. By implication, then, functional projections like IP do not exist in the ILG and abstract agreement and finiteness distinctions are not involved in L2 acquisition.

A more local kind of impairment is proposed by Beck (1998), Eubank (1993/94) and Eubank *et al.* (1997), who argue that feature strength (at least with respect to finiteness) is essentially impaired or inert in the ILG. This was originally assumed by Eubank to be a temporary phase but is now argued to be permanent (Beck, 1998). Optional verb placement is the consequence of inert feature strength: if a feature is neither strong nor weak, a verb can appear in either a raised or an unraised position.

What is not so clear is why Beck and Eubank assume that a verb will be finite when it raises and nonfinite when it does not. Absence of functional projections, features or feature-checking mechanisms in fact implies randomness or mismatches in the use of inflectional morphology (Borer and Rohrbacher, 1997; Lardiere, 2000). As Borer and Rohrbacher (1997: 28) point out:

Without AgrSP, the syntactic appropriateness of fully inflected forms cannot be checked, nor can syntactically inappropriate forms be excluded. A grammar without AgrSP should therefore allow well-formed morphological forms to be distributed randomly, as the mechanism designed to check their appropriateness in a specific syntactic context is non-existent. A similar rationale predicts that without a TP, children should make tense errors, failing to check base-generated tensed forms against the properties of T . . . randomly distributed inflected forms should abound.

We believe that the same rationale applies if TP and AgrP are present but features or feature values are not, that is, if features are impaired or inert as to strength. In such a case, there would be nothing to prevent featural mismatches. As far as we can see, this indeed implies that there should be no contingency between finiteness of the verb and verb placement. Both morphological and syntactic variability should be found; in other words, incidence of finite/non-finite morphology will not be structurally determined.

V Predictions

A number of questions remain unanswered by the research conducted so far. One issue is whether or not variation in L2 morphology is unconstrained, as might be expected on the views that assume impairment, with breakdown in functional categories or features, hence in feature-checking mechanisms. In other words, is it the case that finite and non-finite forms alternate freely, while agreement morphology is often inaccurate? Here, we consider predictions relevant to finiteness and to agreement.

As far as finiteness is concerned, on the MSIH, finite forms are assumed to be genuinely finite, hence they should occur in finite positions but not in non-finite ones. Nonfinite forms, on the other hand, are sometimes truly non-finite and at other times occur as substitutes for finite inflection, hence their position is more variable. Thus, we predict that finite verbs will be found only in finite contexts. In particular, if a verb is finite, it should not be used after a preposition or another verb such as an auxiliary or modal in the same clause. Only non-finite forms should be used in such contexts. Moreover, finite forms should appear before negation in L2 French or German and should not be found following a negator.

The IRH, on the other hand, predicts no clear contingencies between finiteness and verb position: if features or feature strength are lacking, finite verbs should be found in both raised and unraised positions. Thus, the two hypotheses differ as to their predictions for finite verb forms. As far as non-finite forms are concerned, both hypotheses make the same prediction, though for different reasons, namely that they will occur in both finite and non-finite contexts.

As for agreement, where finite forms are used, the MSIH predicts agreement will be appropriate, since the relevant features and feature-checking mechanisms are assumed to be present (cf. Poeppel and Wexler, 1993; Borer and Rohrbacher, 1997). At least the strong form of the IRH (Meisel, 1991; 1997) predicts agreement mismatches, since the mechanisms necessary for checking agreement features are impaired.

VI Current study

In the present article we further examine spontaneous production data obtained longitudinally from four adults learning L2 French and L2 German in naturalistic environments (for background information see Table 1). The two learners of L2 French – Abdelmalek and Zahra – were native speakers of Moroccan Arabic. They were interviewed roughly once a month for a little over three years by the European Science Foundation Project on L2 acquisition by Adult Immigrants (Perdue, 1984; 1993). Both learners had immigrated to France from Morocco, where they had had no previous exposure to French. Their first interviews took place after one year in France, during which time they had little contact with French. At that point, their L2 proficiency was judged to be very limited. As for the two learners of German – Ana and Zita – they were native speakers of Spanish and Portuguese respectively. They were immigrants to Germany and were interviewed by the ZISA project (Clahsen *et al.*, 1983). They were first recorded three months after their arrival in Germany. While neither of them had been exposed to German in their native countries, once in Germany Ana had much more exposure to the language than Zita. Ana also had started taking classes in German one month prior to the first recording session. Both learners were interviewed on average once a month for a little less than two years. (Note that there is a break in Ana’s data between months 14 and 23 as she returned to Spain for 9 months. Recording resumed when she came back to Germany.)

Utterances containing a verb and consisting of at least two constituents were retained for analysis. Items clearly identified as repetitions and formulaic expressions were discarded. Potentially problematic is the fact that both French and German inflectional paradigms contain homophonous morphemes. As can be seen in Table 2, singular forms in French are homophonous in each of the three verb-classes in the present tense.³ Distinctive forms are only found in the plural (but 3rd person plural is homophonous with the

Table 1 Learner and interview details

	Abdelmalek	Zahra	Ana	Zita
L1	Arabic	Arabic	Spanish	Portuguese
L2	French	French	German	German
Age at onset	adult	34	22	17
Number of files	25	25	17	25

³ We gloss forms which are ambiguous because of this homophony as 1/2/3S.

Table 2 French inflectional paradigm (present tense)

		1st group <i>chanter</i> ('sing')	2nd group <i>finir</i> ('finish')	3rd group <i>vendre</i> ('sell')
Infinitival form		chant-er ([e])	fin-ir	vend-re
Past participle		chant-é ([e])	fin-i ([i])	vend-u
Present tense	1/2/3 singular	chant-e/-es/-e	fin-is/-is/-it ([i])	vend-s/-s/-ø
	1/2/3 plural	chant-ons/-ez/-ent	fin-issons/-issez/-issent	vend-ons/-ez/-ent

Table 3 German inflectional paradigm (present tense)

		<i>lernen</i> ('learn')
Infinitival form		lern-en
Past participle		ge-lern-t
Present tense	1/2/3 singular	lern-e/-st/-t
	1/2/3 plural	lern-en/-t/-en

singular in the 1st group, *-ent* being silent). In the case of the 1st group, the 2nd person plural marker *-ez* is homophonous with the infinitival suffix *-er* and past participle marker *-é*. In the case of the 2nd group, singular forms ending in [i] are homophonous with the corresponding past participle. In our investigation, a verb form was considered non-finite in the absence of clear evidence to the contrary.

German, on the other hand, has more distinctive forms in the singular than in the plural (see Table 3). The 1st and 3rd person plural markers are homophonous with each other and with the infinitival marker *-en*.

VII Results

The results reported in this article concern both root and non-root contexts, including matrix declaratives, embedded clauses and questions. We report total number of occurrences of the forms we are interested in, collapsed across all interviews, our rationale being as follows. According to the IRH, functional categories or features are permanently impaired. Therefore, we should find evidence for variable placement of finite verbs, as well as agreement problems, throughout the data, including in the later interviews. The opposite should obtain for the MSIH: since features are assumed to be available in the initial state of L2 acquisition (either from the L1

or from UG),⁴ finite verbs should not be randomly positioned in any of the files being examined, and agreement should be accurate when it occurs. Finite and non-finite verb forms are indeed found in almost all the files examined (Prévost, 1997; Prévost and White, 1999). It is not the case, for example, that non-finite forms occur solely in the earliest interviews and that finite forms emerge later on. (We return to the issue of development in the discussion.)

1 Finite forms in non-finite contexts

a Verbs occurring after auxiliaries, prepositions and other verbs: Table 4 summarizes the incidence of finite and non-finite verb forms in clearly non-finite contexts, namely following a preposition, an auxiliary or another verb, such as a modal.⁵ Considering the data as a whole, the incidence of finite forms in these environments is very low, the vast majority of verbs appropriately being non-finite. In addition, there is no period in the data, not even in the earlier files, during which finite forms are found in non-finite contexts or are randomly distributed in finite and non-finite environments.

Examples of appropriate use of non-finite forms in non-finite contexts are given in (3) through (5). In some cases, the past participle is used instead of the infinitival form (6) and vice versa

Table 4 Finiteness of verbs following auxiliaries, prepositions and other verbs

	L2 French				L2 German			
	Abdelmalek		Zahra		Ana		Zita	
	-fin	+fin	-fin	+fin	-fin	+fin	-fin	+fin
Aux + V	180	3	98	0	12	2	32	1
Prep + V	28	3	36	1	22	1	3	0
V + V	26	8	17	1	30	2	35	1
Total	234	14	151	2	64	5	70	2
		(5.6%)		(1.3%)		(7.2%)		(2.8%)

⁴Proponents of the MSIH include those advocating Full Transfer/Full Access (e.g. Haznedar and Schwartz, 1997) as well as those advocating Full Access to UG without transfer (Epstein *et al.*, 1996). These views agree that the early IL grammar is not impaired or underspecified with respect to functional categories or features, even though they disagree on the source of the initial representation of these categories. For the present purposes, this issue can be left open.

⁵In German root clauses, non-finite verbs used with auxiliaries and modals appear in final position. Here, we will not be concerned with the acquisition of target verb-placement: the focus of the study is whether or not non-finite verbs bear the infinitival marker, not whether they are correctly positioned in the clause. The predictions made by the MSIH and the IRH with respect to morphology are not affected by the issue of verb-placement in German. See also note 7.

(7). Nonfinite verb forms are also found as complements of adjectives, with the preposition omitted (8).⁶

- 3) a. il est parti l'Espagne, lui (Abdelmalek, month 14)
he is gone-PP the Spain him
b. Malika est sortie (Zahra, month 23.7)
Malika is gone+out-PP
c. ich habe gekommen drei mal in Wohnung (Ana, month 11.7)
I have come-PP three times in house
d. meine Schwester hat nich gesehen (Zita, month 25.6)
my sister has not seen-PP
- 4) a. pour voir (Abdelmalek, month 15)
for see-INF
b. sans déclarer (Zahra, month 20)
without declare-INF
c. für nehmen (Ana, month 4)
for take-INF
b. ich hab zu tun (Zita, month 25.8)
I have to do-INF
- 5) a. je te fais manger (Abdelmalek, month 17.7)
I you make eat-INF
b. je veux partir (Zahra, month 21.7)
I want leave-INF
c. er muß zuhause Übungen machen (Ana, month 13.5)
he must at+home practice do-INF
d. ich weiß nich machen (Zita, month 11.7)
I know not do-INF
- 6) a. il faut parti (Abdelmalek, month 27.7)
it must leave-PP
'We must leave.'
b. je peux servi (Zahra, month 17)
I can serve-PP
- 7) a. il a faire (Zahra, month 21.7)
he has do-INF
b. ich habe machen ein Jahre mehr (Ana, month 11.7)
I have do-INF one year more
c. mein Schwager is schon gehen (Zita, month 24.4)
my brother-in-law is already go-INF
- 8) a. c'est difficile chercher un patron (Abdelmalek, month 20.5)
it is difficult look+for-INF a boss
b. c'est pas bon manger à la cuisine (Zahra, month 23.7)
it is not good eat-INF in the kitchen
c. für mich is es egal fahren nach alte Stadt (Ana, month 13.5)
for me is it equal drive+INF to old city

⁶The following abbreviations will be used: INF = infinitive; PP = past participle; S = singular; P = plural; 1 = 1st person; 2 = 2nd person; 3 = 3rd person; SUBJ = subjunctive; MASC = masculine; FEM = feminine.

Many of Abdelmalek's errors involving finite verbs in non-finite contexts occurred with the expression of obligation, 'il (expletive) faut + V', as illustrated in (9). Here, the second verb should be an infinitive but Abdelmalek produces a finite verb.

- 9) a. il faut marche (Abdelmalek, month 36.7)
 it must walk-1/2/3S
 'We must walk.'
 b. il faut paye(s) (Abdelmalek, month 38.7)
 it must pay-1/2/3S
 'You must pay.'

The verb *falloir* in fact subcategorizes for either a non-finite CP (10a) or a finite verb in the subjunctive (10b). Abdelmalek very often omits complementizers in the finite case, producing utterances such as (11a). He also uses a high number of subjectless CPs throughout his interviews, as illustrated in (11b). Thus, it might be the case that the finite verbs following the expression *il faut* in (9) involve a finite CP with both the complementizer and the subject missing: *il faut* [_{CP} Ø [_{IP} Ø marche]].

- 10) a. il faut partir
 it must go-INF
 'We have to leave.'
 b. il faut que nous partions
 it must that we leave-SUBJ+1P
 'We have to leave.'
- 11) a. il faut ø tu partes (Abdelmalek, month 27)
 it must you go-SUBJ+1/2/3S
 'You have to leave.'
 b. parce que ø peut pas donner (Abdelmalek, month 17.7)
 because can-1/2/3S not give-INF

As for L2 German, most finite verbs which incorrectly appear in non-finite contexts either bear the 1st person singular *-e* marker (12) or appear as bare stems in the case of Ana (13). The verbs may in fact be non-finite in both cases (for L2 acquisition, see Meisel, 1991; 1997; for L1 acquisition, see Boser *et al.*, 1992). In particular, Meisel points out that *-e* forms can be regional variants of verbs ending with the infinitival marker *-en*. (For further discussion see the section 'Verbal agreement' on page 120.)

- 12) a. für lerne Deutsch is nicht gut (Ana, month 13.5)
 to learn-1S German is not good
 b. du willst nich arbeite hier (Zita, month 24.4)
 you want not work-1S here

- 13) ich will diese Jahre fahr nach Spanien (Ana, month 13.5)
 I want this year drive-Ø to Spain
 mit mein Eltern
 with my parents

b Verb placement and negation: A standard test for finiteness in languages like French and German involves verb placement with respect to the negative. Finite verbs have strong features and raise over negation, whereas non-finite verbs do not; see (1) above. According to the IRH, feature strength determining verb-placement is impaired. In consequence, finite and non-finite forms should be found on either side of the negative *pas* in French or *nicht* in German. In contrast, the MSIH holds that abstract feature strength is unimpaired; hence, finite verbs should systematically precede the negator in L2 French and L2 German.⁷ The results confirm this prediction, as shown in Table 5. The systematic placement of finite verbs before the negator suggests that verb-movement has occurred and, hence, that the features and feature strength involved in raising (and checking) are known to the learners. In addition, correct verb-placement as regards finite verbs is observed from the very first instances of negative clauses, which leads us to conclude that appropriate feature strength was posited early on. Examples of correct negative placement are presented in (14) and (15).

- 14) a. mais on peut pas dormir (Abdelmalek, month 17.7)
 but one can-1/2/3S not sleep-INF
 b. i mange pas (Zahra, month 26.7)
 he eat-1/2/3S not
- 15) a. ich studiere nicht (Zita, month 3.7)
 I study-1S not
 b. ich spreche nicht Deutsch (Ana, month 4.5)
 I speak-1S not German

Table 5 Verb placement with respect to negation

	L2 French				L2 German			
	Abdelmalek		Zahra		Ana		Zita	
	V-Neg	Neg-V	V-Neg	Neg-V	V-Neg	Neg-V	V-Neg	Neg-V
Finite	90	3	135	0	82	2	74	4
Non-finite	6	44	7	5	9	12	13	29

⁷ In German embedded clauses, the negator precedes the clause-final finite verb. However, the L2 German data reviewed here suggest that the two learners of German (whose L1s are SVO) have not acquired the V-final property of German (see Prévost, 1997). Negative embedded clauses are thus treated on a par with negative root clauses in which the finite verb raises above negation.

As for non-finite forms, Table 5 reveals that there is variability in verb placement. With the exception of Abdelmalek, whose non-finite forms occur almost exclusively after the negator, the other subjects show non-finite verb forms in raised and unraised positions.⁸ (Similar variability is reported by Müller (1998) in a case study of Bruno, an adult learner of L2 German.) We assume that non-finite forms occurring after the negative are genuinely non-finite: verb movement does not take place, hence the verb appears after the negator, giving Neg V order. However, when non-finite forms are used as substitutes for finite ones, verb-raising occurs, yielding the order V Neg.

2 *Overuse of finiteness and non-finiteness compared*

In Prévost and White (1999), we report incidence of non-finite verb forms in a number of finite contexts. Table 6 summarizes those results from three contexts involving the projection of Infl:

- 1) in CPs, such as questions and subordinate clauses, the projection of C implies Infl;
- 2) DP subjects must appear in Spec of IP for case assignment/checking reasons;
- 3) clitic subjects typically attach to the verb in Infl.

If Infl is projected, verb-movement will occur for feature-checking purposes, in which case the verb should be finite and should bear

Table 6 Incidence of non-finite verbs in finite contexts (as a percentage of all verbs in finite contexts)

	L2 French		L2 German	
	Abdelmalek	Zahra	Ana	Zita
CPs with non-finite verbs	54/146 (37%)	65/222 (29.3%)	13/137 (9.5%)	18/82 (22%)
DP subjects with root infinitives	15/68 (22%)	32/103 (31.1%)	23/206 (11.2%)	54/250 (21.6%)
Subject clitics with root infinitives	168/700 (24%)	120/512 (23.4%)		
Total	237/914 (25.9%)	217/837 (25.9%)	36/343 (10.5%)	72/332 (21.7%)

⁸The figures presented in Table 5 differ slightly from what is reported in Prévost and White (1999) (their Table 7). This is due to the fact that in the present study we looked at both root and non-root contexts while the negation data in Prévost and White (1999) were restricted to root declaratives. Here, we also consider negative clauses involving a complex tense (e.g. Aux-Neg-V_{-fin}) as instances of both V_{+fin}-Neg and Neg-V_{-fin}. Negative clauses whose main verb was a bare stem other than a modal or a first person singular form in L2 German were discarded.

Table 7 Overuse of non-finiteness vs. overuse of finiteness

		Obligatory finite contexts		Obligatory non-finite contexts	
		+finite	–finite	–finite	+finite
L2 French	Abdelmalek	767	243 (24.1%)	278	17 (5.8%)
	Zahra	755	224 (22.9%)	156	2 (1.3%)
L2 German	Ana	389	45 (10.4%)	76	7 (8.4%)
	Zita	434	85 (16.4%)	98	6 (5.8%)

inflectional morphology. In the data we examined, however, the verb form was sometimes non-finite, as can be seen in Table 6.

If non-finite forms occurring in finite positions are in fact default substitutes for finite forms, then one expects the overuse of non-finite forms to be higher than the overuse of finite forms. Table 7 compares the incidence of non-finite forms in finite positions (from Tables 5 and 6) to incidence of finite forms in non-finite positions (from Tables 4 and 5). As Table 7 demonstrates, the incidence of non-finite verbs in finite contexts is much higher than the incidence of finite verbs in non-finite contexts for 3 of the 4 subjects. In other words, while non-finite forms occur in both non-finite and finite positions, finite forms are largely restricted to finite (raised) positions. The contingency between finiteness and context is significant for all learners but Ana, who was the most advanced (Abdelmalek: $\chi^2 = 47.91$, $p < .0001$; Zahra: $\chi^2 = 39.909$, $p < .0001$; Zita: $\chi^2 = 7.817$, $p < .01$; Ana: $\chi^2 = .288$, $p > .05$). Such a contingency is problematic for the IRH, since positioning of verbal forms should not depend on finiteness if feature strength is impaired. In contrast, results are consistent with the MSIH, which assumes that syntactic consequences of finiteness are known to learners. As a result, finite verb forms are associated with finite features and appear in raised positions. Nonfinite forms appear correctly in non-finite contexts and also as a default in finite positions.

3 Accuracy on agreement

The MSIH predicts that when agreement is found, it should be used correctly since agreement features and feature-checking mechanisms are assumed to be unimpaired. The IRH (at least, Meisel's version) makes the opposite prediction: erroneous agreement should be found, since checking cannot take place and there is no way of excluding agreement mismatches. Here we consider two kinds of agreement: verbal agreement in French and German and clitic agreement in French clitic-doubling

constructions, on the assumption that subject clitics are agreement markers which appear in Infl (Auger, 1994).

a Verbal agreement: We investigated knowledge of agreement by identifying inflected verbs. In each case, we looked at person and number inflection on the verb and then at whether the subject of that verb was appropriate, given the inflectional morphology. In other words, following Poeppel and Wexler (1993), we look at whether agreement is accurate when it is present and not at whether a subject is found with an agreeing verb form (contra Meisel, 1991). In addition to the regular inflectional paradigms presented in Tables 2 and 3, both languages include verbs with irregular paradigms involving suppletion (e.g. *être* ('be'), *avoir* ('have'), *aller* ('go') in French and *sein* ('be') in German). Suppletive forms have traditionally been excluded from the investigation of knowledge of agreement and from the study of the relationship between morphological and syntactic knowledge. Yet, as Lardiere (2000) reminds us, suppletive forms are endowed with features that need to be checked. They are thus indicative of knowledge of agreement and it is appropriate to examine accuracy in production of suppletive forms. (Furthermore, suppletive forms avoid the problem of homophony which arises with regular verbs, since almost all persons are realized with a distinct form.)

As can be seen in Table 8, agreement is largely correct in the L2 French data. Accurate use of inflectional marking is around 95%.⁹ While results on regular affixation are difficult to interpret, since many agreement markers are homophonous, it is noteworthy that

Table 8 Accuracy in use of verbal agreement (L2 French)

	Abdelmalek		Zahra	
	Correct	Incorrect	Correct	Incorrect
Regular affixation	447 (94.7%)	25	552 (93.4%)	39
homophonous inflection	445 (94.7%)	25	538 (93.2%)	39
distinctive inflection	2 (100%)	0	14 (100%)	0
<i>être, avoir</i> and <i>aller</i> forms	264 (97.8%)	6	156 (98.7%)	2
Total	711 (95.8%)	31	708 (94.5%)	41

⁹ Some inflected forms involving the masculine 3rd person plural clitic *ils* ('they') were excluded from analysis since the clitic is homophonous with the corresponding singular form *il* ('him'). Thus, there is no way to tell whether the verb form is appropriate or not, as shown in (i), where the verb is 3rd person singular and the clitic is ambiguous between singular and plural. The NP that the clitic doubles is plural.

i) les autres i est parti (Abdelmalek, month 34.5)
the others they/he(?) is-3S gone

accuracy on the distinct suppletive forms of *être*, *avoir* and *aller* is almost 98%.

Examples of appropriate person/number agreement are shown in (16). Errors found in the regular paradigm mostly involved verbs of the 3rd group, pronounced incorrectly with a final [t] or [d], as in (17a). All Abdelmalek's errors and 90% of Zahra's errors were of this type. In French, a root-final [t] or [d] is pronounced in the singular form verbs of the 1st group in the present tense (see Table 2). (For example, *je/tu/il chante(s)* ([sât] 'I/you/he sing(s)') and *je/tu/il garde(s)* ([ga d] 'I/you/he keep(s)').) It is thus possible that the learners were overgeneralizing this regular inflection pattern to verbs of the 3rd group like *partir* ('go') and *vendre* ('sell') where root-final [t] or [d] are normally silent. As for the few incorrect suppletive forms, most of them are person errors – as shown in (17b) and (17c) – involving overgeneralization of 3rd person singular forms, such as *a* ('has') and *est* ('is').

- 16) a. il donne une bière (Abdelmalek, month 14)
 he give-1/2/3S a beer
 b. moi j'aime pas de monde (Abdelmalek, month 25)
 me I like-1/2/3S not some people
 c. elle (l') a donné le prof (Zahra, month 12)
 she (it) have-3S given the teacher
 d. quand le matin ils sortent le travail (Zahra, month 29.2)
 when the morning they leave-3P¹⁰ the work
- 17) a. moi parte (Zahra, month 15.5)
 me go-1/2/3S
 b. j'a fait la bagarre avec lui-même (Abdelmalek,
 I have-3S made the fight with him month 52.7)
 c. quand moi est pas là (Zahra, month 39.5)
 when me is-3S not there

In L2 German, accuracy is also very high, as can be seen in Table 9 (for similar observations, see also Clahsen, 1988). This is particularly obvious as far as suppletive forms are concerned where accuracy is comparable to that in L2 French. Results on regular affixation were expected to be somewhat more revealing in L2 German since there are more distinct morphological forms in German than in French. However, what we found was relatively mixed: accuracy is very high on 3rd person singular *-t* for both Zita and Ana; Ana shows total accuracy on 2nd person singular *-st*, while Zita is somewhat less accurate; both learners show lowest accuracy

¹⁰ Given the fact that these learners pronounce the final consonant in singular forms of verbs of the 3rd group, this form could be 1/2/3S, even though this would not be the case for a native speaker of French.

Table 9 Accuracy in use of verbal agreement (L2 German)

		Ana		Zita	
		Correct	Incorrect	Correct	Incorrect
Regular affixation	1S (-e)	155 (74.5%)	53	70 (70.7%)	29
	2S (-st)	17 (100%)	0	33 (80.5%)	8
	3S (-t)	46 (95.8%)	2	90 (90.9%)	9
	Total	218 (79.8%)	55	193 (80.7%)	46
<i>sein</i> forms	1S (<i>bin</i>)	19 (100%)	0	12 (100%)	0
	2S (<i>bist</i>)	0	0	2 (100%)	0
	3S (<i>ist</i>)	230 (93.9%)	15	111 (97.4%)	3
	3S-Past (<i>war</i>)	5 (100%)	0	0	0
	1P (<i>sind</i>)	16 (100%)	0	0	0
	3P (<i>sind</i>)	28 (87.5%)	4	2 (100%)	0
	Total	298 (94%)	19	127 (97.7%)	3
<i>kann/will/muß</i> stems	1/3S	45 (95.7%)	2	48 (96%)	2
Total		561 (88.1%)	76	368 (87.8%)	51

(albeit higher than chance) on *-e*. Finally, accuracy is very high on modals such as *können* ('can'), *müssen* ('must') and *wollen* ('want') which do not bear suffixes in the 1st and 3rd person singular. (Müller (1998) reports that Bruno also shows high accuracy of agreement with *-t* and *-st* and lower with *-e*.)

Examples of correct agreement are given in (18) and incorrect agreement in (19).

- 18)

a.

er sagt der Herr

he say-3S the man

b.

ich lese wenig

I read-1S little

c.

wir lernen

we learn-1P

d.

hunderte du bekommst

hundred you get-2S
- (Ana, month 4)

(Ana, month 7.4)

(Zita, month 6.5)

(Zita, month 15)
- 19)

a.

er kaufe ein Blume

he buy-1S a flower

b.

Michael spiele hier

Michael play-1S here

c.

die Männer arbeite ganze Tag

the men work-1S whole day
- (Ana, month 4)

(Zita, month 10)

(Zita, month 23.5)

The relatively high number of erroneous *-e* forms in the data are, on the face of it, somewhat problematic for our hypothesis. These forms co-occur with cases of *-e* appropriately agreeing with a 1st person subject (over 70% showing appropriate agreement). Most of the errors involve a 3rd person singular subject, as in (19a) and

(19b). One possibility is that the *-e* forms are non-finite, as discussed earlier and assumed by Meisel (1991). If these forms are non-finite, then they simply constitute further instances of missing surface inflection, with a non-finite form being used in place of a finite one. However, in that case one might expect a higher incidence of verbs ending in *-e* in genuinely non-finite positions. Ana produced only 3 *-e* forms following a preposition or another verb (compared to 64 verbs ending in *-en*), while Zita produced only one such form compared to 70 *-en* forms (see Table 4 above). However, Müller (1998: 136) reports that Bruno does use *-e* in non-finite contexts, including after prepositions and modals. Sixteen per cent of his verb forms occurring after modals end with *-e*. Another possibility is that *-e* represents some kind of default agreement.

Turning now to the other errors, Zita's incorrect usage of *-st* is largely lexically-driven: half of her errors involve the verb *sagen* ('say'), as illustrated in (20a). Ana's errors on *ist* ('is') and *will/kann/muß* ('want/can/must') occur in plural contexts – (20b and 20c) – often involving the subject *die Leute* ('the people') which is singular in Spanish, Ana's mother tongue (*la gente*). This might explain her use of singular agreement in this case (see also Müller, 1998).

- 20) a. mein Schwager sagst auch (Zita, month 22)
 my brother-in-law say-2S too
 b. aber die Leute ist ein bißchen müde (Ana, month 13.5)
 but the people is-3S a little tired
 c. viele Leute will nich <commenzar> <estudiar> (Ana, month
 many people want-ø not start study 4.5)

In general, our results differ from Meisel (1991), who reports generally low accuracy in verbal inflection. However, his study and ours contrast in the method used to assess knowledge of agreement. Meisel first looked for subjects and then checked whether the verb showed appropriate agreement with the subject. He found that many verbs did not agree with the subject, either displaying the non-finite marker *-en* or not being inflected at all, which led him to conclude that knowledge of agreement was impaired. In contrast, we agree with Poeppel and Wexler (1993) that the most appropriate way to determine knowledge of agreement is to look at whether a form, if inflected, shows accurate agreement. Our results suggest that agreement is in fact in place: when an inflected form is used, agreement is largely accurate (see also Clahsen, 1988). The claim of the MSIH is that the non-finite or bare forms that occur are in fact finite defaults, rather than evidence of incorrect agreement.

b Subject clitics (L2 French): The investigation of subject clitics is restricted to L2 French since German does not have such elements. Assuming that subject clitics are agreement markers appearing in Infl (Auger, 1994), the MSIH predicts that when they occur in clitic-doubling constructions, they should agree with the co-referring subject NP. Under the IRH, such agreement is not expected since agreement features are held to be impaired. Results in Table 10 show that subject clitics agree in person, number and gender with the NP to a large extent, in support of the MSIH.

Examples in (21) show subject clitics agreeing with the NP in both finite and non-finite environments.¹¹ In (21d), the gender assigned to the NP is masculine instead of feminine, but the agreeing clitic is also masculine.

- 21) a. *il est parti l' Espagne, lui* (Abdelmalek, month 14)
he is-3S gone the Spain him
b. *moi j'ai pas de quarante francs* (Abdelmalek, month 14)
me I have-IS not some forty francs
c. *monsieur il arriver* (Zahra, month 18.5)
mister he arrive-INF
d. *le fille il a volé le pain* (Zahra, month 27.7)
the-MASC girl he has-3S stolen the bread

Most errors involved the masculine 3rd person singular clitic *il* ('he'), which appears to be used as yet another default form. Erroneous usage of *il* involves gender (where it replaces the corresponding clitic *elle*), as in (22), or person, as in (23). Such errors constitute 95% of Abdelmalek's clitic errors and 78.5% of Zahra's.

Table 10 Subject clitics in clitic doubling constructions

	Abdelmalek		Zahra	
	Clitic agrees with subject	Clitic does not agree with subject	Clitic agrees with subject	Clitic does not agree with subject
Finite verb forms	103	17	111	8
Non-finite verb forms	17	2	30	6
Total	120	19 (13.5%)	141	14 (9%)

¹¹What is at stake here is the agreement between the clitic and the NP, regardless of the form of the verb. In Prévost and White (1999), we present data showing that both finite and non-finite verb forms are found with subject clitics in the case of adult learners. We argue that the incidence of non-finite verbs with clitics is supportive of the MSIH. See also Table 6.

- 22) a. la sécurité sociale, il paye (Abdelmalek, month 51.5)
the-FEM security social he pay-1/2/3S
b. la dame i travaille avec son mari (Zahra, month
the-FEM woman he work-1/2/3S with her husband 26.7)
- 23) a. toi il a raison (Abdelmalek, month 24)
you he has-3S right
b. moi aussi il parle avec la dame (Zahra, month 41)
me too he speak-1/2/3S with the woman

To summarize the results on agreement, in the case of verbs in L2 French and German as well as subject clitics in L2 French, agreement is largely accurate. When an inflectional suffix bearing agreement is found on a verb, it is appropriate for the subject in question; when a clitic is found, it is appropriate for the NP that it doubles. There are two main exceptions to this observation:

- 1) in L2 German, the 1st person singular ending *-e* is found with subjects that are not 1st person singular;
- 2) in L2 French, the 3rd person masculine singular clitic *il* is found with subjects that are not 3rd person masculine.

We suggest that these may be default agreement markers.

VIII Discussion and conclusion

Both hypotheses discussed in this article assume variability in incidence of inflected verb forms in L2 acquisition. The MSIH and the IRH make similar predictions about non-finite morphology, assuming that non-finite forms will be found in finite contexts. In the case of the MSIH, this is because non-finite forms are resorted to as a default (in a sense to be made clearer below). Although non-finite in form, they behave syntactically like finite verbs. In the case of the IRH, on the other hand, incidence of non-finite forms in finite contexts is due to a breakdown in feature-checking mechanisms, with the consequence that non-finite forms can occur anywhere.

Where the two hypotheses differ is over their claims about finite morphology, as well as agreement, and this is what we have concentrated on in the present article. Our L2 French and German data show that, with a few exceptions, finite forms do not occur in non-finite contexts: they are not found after a preposition or a negator, and they are not used along with another verb in the same clause. These results suggest that adult learners are in fact distinguishing between finite and non-finite morphology and that finite forms do not substitute for non-finite. Finite forms rarely

occur in low positions, suggesting that finite features are present in the ILG rather than being impaired. In addition, we find little evidence of 'faulty' agreement, in the sense of mismatches (or feature clashes) between person, number and gender inflection on a verb/clitic and the corresponding subject. In general, then, the results support the MSIH.

By contrast, the data fail to confirm the IRH. We have argued that the implication of missing functional categories or features (Meisel, 1997) and inert feature strength (Eubank *et al.* 1997; Beck, 1998) is that finite verb forms should be relatively variable. If features are lacking in IL grammars, feature-checking mechanisms cannot apply and finite forms should appear in any position, finite or non-finite. If feature strength is impaired, there is no motivation for finite verbs to raise systematically; hence, they may remain within the VP. Thus, the strong contingency between verb placement and finiteness uncovered in this study is problematic for the IRH (*pace* Eubank and Beck, 1998). Meisel's approach further predicts a high incidence of faulty agreement since all features, including phi-features, are held to be missing. Again, this is not supported by the data. Eubank and Beck do not make the same predictions as Meisel on accuracy of agreement, since only V-features related to finiteness are held to be inert. However, it is not clear under their approach why certain features should be subject to such inertness and others not.

The IRH predicts greater variability involving finiteness than the MSIH, in other words, finite forms occurring in non-finite positions and non-finite forms in finite positions. In contrast, the variability assumed by the MSIH is more constrained; variability is limited to (apparently) non-finite forms which may be found in finite contexts. In that respect, the MSIH makes a clear distinction between knowledge of surface morphology and knowledge of the abstract features underlying movement (see Clahsen (1988) who also argues for a dissociation between knowledge of abstract agreement and surface morphology). Though the L2 learners investigated here produced main verbs with infinitival markers, they also showed knowledge of finiteness and its syntactic consequences with respect to negation, supporting the claim that they have set L2 feature strength to [+strong].

It is of course possible for proponents of the IRH to argue that although abstract functional categories, features or feature values are impaired, L2 learners can be relatively successful in the domain of morphology by resorting to 'parasites' which patch up deficient aspects of the grammar (Beck, 1997) or to explicitly learned inflectional paradigms (Bley-Vroman, personal communication;

Meisel, personal communication). In that case, apparently successful L2 acquisition of morphology tells one nothing, being taken to indicate successful ‘parasites’ rather than appropriate knowledge of both abstract features and surface morphological realization. Lack of success, on the other hand, is more revealing. On the IRH (assuming parasites or explicit learning), it is not clear why, when mistakes are made, they should be in one direction rather than another.

We have argued that L2 learners have abstract features for finiteness and agreement in their interlanguage representation, as evidenced by the syntactic and morphological behaviour of finite verbs. They do, however, exhibit problems with the surface morphological realization of particular forms, sometimes resorting to default forms; in other words, there are what Lardiere (2000) calls ‘mapping problems’ between surface forms and abstract features. This raises the question of what mechanisms underlie the appearance of defaults. In other words, what is their featural content? One possible solution is provided by Distributed Morphology (DM) (Halle and Marantz, 1993), which makes a clear distinction between grammatical features (such as tense, person, etc.) on a given inflected form and the node that hosts it in the syntax.¹²

In DM, each inflected form is assumed to be associated with a bundle of features, such as tense, person, number and gender. For lexical insertion to take place, the features of the vocabulary item must be consistent with the features of the terminal node in the syntax where it is to be inserted. While the features of a syntactic node will be fully specified, features of a lexical item may be partially specified – in that some features may be lacking – or underspecified – in that some feature values may not be specified (Lumsden and Halefom, 1998). The features of the lexical item do not need to exactly match all the features of the hosting node: it is sufficient that they form a proper subset of the feature bundle of that node. In the absence of an exact match, there is a competition between potential candidates for insertion, the winner being the form with the most features that match those of the terminal node.

Our assumption is that L2 learners have acquired the relevant features of the terminal nodes in the syntax (from the L1, from UG or motivated by L2 input). Their problems lie with the feature specification of the associated lexical items. In the grammar of an adult native speaker, non-finite forms are specified as [–finite] and

¹² See Lumsden and Halefom (1998) for a development of DM which is consistent with minimalist checking theory, dispensing with Halle and Marantz’ distinct level of morphological structure.

finite ones are [+finite]. In addition, finite forms are associated with [$\pm$ past] and phi-features (person, number and gender). Non-finite forms cannot be inserted in any position where finite inflection is required because of mismatching features; the same is true of non-agreeing forms. Now suppose that in adult L2 acquisition, non-finite forms are underspecified with respect to finiteness (i.e. they are α finite). If so, then non-finite forms may be inserted into a node involving the feature [+finite]. This would explain the appearance of non-finite forms in finite positions. Finite forms, on the other hand, are not underspecified, hence cannot be inserted into non-finite positions. The same argument applies to the German *-e* form and the French clitic *il*, which we assume to be underspecified for agreement features (i.e. they are α person, α number, α gender), yielding agreement mismatches with the subject (for related proposals, see Müller, 1998). In all three cases, insertion may take place because there is no conflict between what is required by the terminal node and what is specified on the verb form in question.

A related proposal is made for L1 acquisition by Ferdinand (1996). She argues that in the L1 acquisition of French, the 3rd person singular form serves as a finite 'elsewhere' form (in our terms, a default) which is not specified for person or number. As a result, it shows up with 1st and 2nd person singular subjects and with plural subjects. Initially, only the elsewhere form is found. This is followed by a stage where the more specified forms are acquired but the elsewhere form is still used. Finally, only fully specified verb forms are found. Overgeneralizations of elsewhere forms are only in one direction: while a 3rd person singular verb may occur with subjects that are not 3rd person singular, plural verb forms do not occur with singular subjects, nor do 1st person verb forms occur with 3rd person subjects, suggesting that these are fully specified for relevant featural content. In addition to elsewhere forms on verbs, Ferdinand proposes that L1 acquirers have an elsewhere form of subject clitics, underspecified for agreement features. In L1 acquisition, this takes the form of a reduced vowel, whereas in our L2 data, the 3rd person singular clitic *il* acts as a default form, as we have seen.

Acquisition of L1 morphology is always successful (except in pathological cases). Once morphosyntactic and semantic features have been mapped to a particular form, that association is permanent, preventing non-finite forms from surfacing in finite positions, as well as agreement mismatches (except for the short period of variability between elsewhere forms and agreeing forms noted by Ferdinand). In other words, once the more fully specified forms are acquired, they replace underspecified forms, which can

no longer surface. Hence, one sees a developmental progression in L1 acquisition. The same is only partially true of L2 acquisition. In the case of default agreement, there was in fact some evidence of a developmental progression: in Ana's case, for example, 3rd person singular forms were delayed. She did not produce *-t* forms consistently until month 11.7. Before that, many verbs ending in *-e* were found with 3rd person singular subjects. This was followed by a period of overlap between *-t* forms and *-e* forms. Finally, the number of errors on *-e* went down while *-t* forms were produced more regularly (see also Clahsen, 1988). Müller (1998) also reports that Bruno shows a developmental progression, with forms underspecified for agreement subsequently replaced by forms which are fully specified. So far, then, incidence of defaults in L1 and L2 acquisition appears to be parallel, involving the use of underspecified or default forms before the appropriate agreeing forms have been acquired.

Despite these similarities, there are also considerable differences. Our data show that non-finite forms alternate with finite forms throughout the period of observation in both L2 French and L2 German and that non-finite forms never disappear (for details, see Prévost and White, 1999). Furthermore, Lardiere (1998a; 1998b) shows that tense and agreement morphology remain variable in an end state L2 English grammar. What seems to be going on, then, is that even when more fully specified forms are acquired, they do not necessarily always 'win' in the competition for lexical insertion, so that underspecified forms continue to surface. This suggests that access to the more fully specified lexical entries is sometimes blocked. That is, even when a form specified for the relevant features has been acquired, it becomes temporarily irretrievable from the lexicon. We speculate that this might be due to processing reasons or to communication pressure, in which case one might expect the problem to affect different kinds of language use differentially. For example, L2 learners might be expected to perform more accurately on an untimed grammaticality judgement task (where they have time to access the relevant representation) than in spontaneous production or in timed tasks.

In conclusion, we have argued that L2 learners' treatment of finite morphology is not arbitrary: when a verb is finite, the relevant syntactic reflexes are found, as well as appropriate agreement. We have suggested that problems of adult L2 learners relate to the mapping of specific morphological forms to abstract categories. We do not, however, wish to imply that the problem is trivial. Clearly, there is a real difference between L1 and L2 acquirers in this domain: normal L1 acquirers always acquire the appropriate

inflectional morphology of their mother tongue and use it consistently, whereas L2 learners often do not. However, despite lack of consistent use of verbal morphology, when it occurs, it is systematic, suggesting that there is no impairment at an abstract level.

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